Common Linux commands

sudo apt-get update

```
najneen@REJWANA:~$ sudo apt-get update
[sudo] password for najneen:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [748 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1013 kB]
```

sudo apt-get upgrade

```
najneen@REJWANA:~$ sudo apt-get upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required
  libllvm17t64
Use 'sudo apt autoremove' to remove it.
The following packages will be upgraded:
  gtk-update-icon-cache libgtk-3-0t64 libgtk-3-bin libgtk-3-common linux-li
```

lsb_release -a

```
najneen@REJWANA:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.2 LTS
Release:        24.04
Codename:       noble
najneen@REJWANA:~$
```

df -h

```
najneen@REJWANA:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            1.9G     0  1.9G   0% /usr/lib/modules/5.15.167.4-microsoft-standard-WSL2
none            1.9G  4.0K  1.9G   1% /mnt/wsl
drivers         111G   87G   24G  79% /usr/lib/wsl/drivers
/dev/sdc       1007G  4.4G  952G   1% /
none            1.9G   80K  1.9G   1% /mnt/wslg
none            1.9G     0  1.9G   0% /usr/lib/wsl/lib
rootfs          1.9G  2.4M  1.9G   1% /init
none            1.9G  492K  1.9G   1% /run
none            1.9G     0  1.9G   0% /run/lock
none            1.9G     0  1.9G   0% /run/shm
tmpfs           4.0M     0  4.0M   0% /sys/fs/cgroup
none            1.9G   76K  1.9G   1% /mnt/wslg/versions.txt
none            1.9G   76K  1.9G   1% /mnt/wslg/doc
C:\             111G   87G   24G  79% /mnt/c
D:\             200G  136G   65G  68% /mnt/d
E:\             200G  154G   47G  77% /mnt/e
F:\             200G  179G   22G  90% /mnt/f
G:\             332G  114G  218G  35% /mnt/g
tmpfs           386M   16K  386M   1% /run/user/1000
```

This virtual disk appears to WSL as a full-blown 1 TB+ drive (1007G) — but it doesn't mean your C: drive has 1 TB free. The free space in C drive is just 24G and WSL can use only 24G free space.
WSL2 uses dynamic disk sizing — it pretends you have 1007G, but in reality, it only uses the actual space needed.

free -h

```
najneen@REJWANA:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           3.8Gi       598Mi       3.2Gi       3.1Mi       217Mi       3.2Gi
Swap:          1.0Gi          0B       1.0Gi
najneen@REJWANA:~$
```

System has 3.8 GB of RAM available to WSL

598 MB is currently in use by running applications.

3.2 GB is not being used at all.

1.0 GiB of swap (virtual memory) is available (like backup memory if RAM runs out).
0B used means it's not currently being used (which is good — it means your RAM is enough for now).

Check Disk Health

sudo apt-get install smartmontools

```
najneen@REJWANA:~$ sudo apt-get install smartmontools  # On Debian-based systems
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm17t64
Use 'sudo apt autoremove' to remove it.
Suggested packages:
  gsmartcontrol smart-notifier mailx | mailutils
The following NEW packages will be installed:
  smartmontools
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 643 kB of archives.
After this operation, 2211 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 smartmontools amd64 7.4-2build1 [643 kB]
Fetched 643 kB in 2s (283 kB/s)
```

sudo smartctl -a /dev/sda

```
najneen@REJWANA:~$ sudo smartctl -a /dev/sda
smartctl 7.4 2023-08-01 r5530 [x86_64-linux-5.15.167.4-microsoft-standard-WSL2] (local build)
Copyright (C) 2002-23, Bruce Allen, Christian Franke, www.smartmontools.org

=== START OF INFORMATION SECTION ===
Vendor:               Msft
Product:              Virtual Disk
Revision:             1.0
Compliance:           SPC-3
User Capacity:        407,298,048 bytes [407 MB]
Logical block size:   512 bytes
LU is thin provisioned, LBPRZ=0
>> Terminate command early due to bad response to IEC mode page
A mandatory SMART command failed: exiting. To continue, add one or more '-T permissive' options.
```

Detailed hardware info

sudo lshw -C display

```
najneen@REJWANA:~$ sudo lshw -C display
  *-display
       description: 3D controller
       product: Basic Render Driver
       vendor: Microsoft Corporation
       physical id: 0
       bus info: pci@9c41:00:00.0
       version: 00
       width: 32 bits
       clock: 33MHz
       capabilities: bus_master cap_list
       configuration: driver=dxgkrnl latency=0
       resources: irq:0
```

GPU installation check

nvidia-smi

```
najneen@REJWANA:~$ nvidia-smi
Command 'nvidia-smi' not found, but can be installed with:
sudo apt install nvidia-utils-525         # version 525.147.05-0ubuntu1, or
sudo apt install nvidia-utils-525-server  # version 525.147.05-0ubuntu1
sudo apt install nvidia-utils-470         # version 470.256.02-0ubuntu0.24.04.1
sudo apt install nvidia-utils-470-server  # version 470.256.02-0ubuntu0.24.04.1
sudo apt install nvidia-utils-535         # version 535.183.01-0ubuntu0.24.04.1
sudo apt install nvidia-utils-535-server  # version 535.230.02-0ubuntu0.24.04.3
sudo apt install nvidia-utils-550         # version 550.120-0ubuntu0.24.04.1
sudo apt install nvidia-utils-570-server  # version 570.86.15-0ubuntu0.24.04.4
sudo apt install nvidia-utils-565-server  # version 565.57.01-0ubuntu0.24.04.3
sudo apt install nvidia-utils-550-server  # version 550.144.03-0ubuntu0.24.04.1
najneen@REJWANA:~$ nvidia-smi
```

top

```
najneen@REJWANA:~$ top
top - 04:38:32 up 41 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3856.0 total,   3322.7 free,    568.8 used,    157.8 buff/cache
MiB Swap:   1024.0 total,   1024.0 free,      0.0 used.   3287.2 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   22096  13300   9564 S   0.0   0.3   0:01.10 systemd
      2 root      20   0    2776   1924   1796 S   0.0   0.0   0:00.00 init-systemd(Ub
     10 root      20   0    2776    132    132 S   0.0   0.0   0:00.00 init
     64 root      19  -1   50444  16416  15316 S   0.0   0.4   0:00.23 systemd-journal
    107 root      20   0   23984   6048   4896 S   0.0   0.2   0:00.13 systemd-udevd
    119 systemd+  20   0   21452  11860   9668 S   0.0   0.3   0:00.14 systemd-resolve
    120 systemd+  20   0   91020   6440   5592 S   0.0   0.2   0:00.16 systemd-timesyn
    167 root      20   0    4236   2680   2448 S   0.0   0.1   0:00.01 cron
    168 message+  20   0    9748   5376   4476 S   0.0   0.1   0:00.17 dbus-daemon
    175 root      20   0   17976   8236   7212 S   0.0   0.2   0:00.10 systemd-logind
    178 root      20   0 1756096  16064   9480 S   0.0   0.4   0:00.15 wsl-pro-service
    183 root      20   0    3160   1164   1072 S   0.0   0.0   0:00.01 agetty
    187 root      20   0    3116   1180   1092 S   0.0   0.0   0:00.00 agetty
```